ARAB YOUTH SPACE HACKATHON 2026 · 813 CHALLENGE

هَاكَاثُون الشَّباب العَرَبِيّ
THE ARAB YOUTH HACKATHON

See Beyond Green

Plot-level crop screening in the Souf oasis, El Oued, Algeria, from open Sentinel-2 imagery

Theme: Precision Agriculture and Crop Intelligence

Team Astra

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The problem

1

Seen too late

A failing crop is obvious on the ground only after the loss has happened.

2

Too many plots

The Souf grows about a third of Algeria's potatoes on thousands of small irrigated pivots. One extension officer cannot visit them all.

3

Two crop calendars

Late and main seasons run in parallel, so a normal December harvest looks like a failure to a simple rule.

Our question: which plots should be inspected first?

User: an agricultural extension officer or farm manager. **Decision:** which plots to inspect first this week.

Compare each crop only with crops that emerged in the same weeks

Free Sentinel-2 imagery. No field boundaries and no ground data needed.

1

Detect

Pixels that turn green at least once in the year are cultivated. Connected pixels form a plot.

2

Measure

The green-up and decline dates of each plot give the length of its green season.

3

Flag

A plot is a priority if its season is 25 days or more shorter than that of crops that emerged within 20 days of it.

Output: a ranked inspection list and a map for the extension officer.

Data and study area

≈ 120 km²

Souf oasis, El Oued, Algeria (6.903 to 7.019 E, 33.422 to 33.523 N)

503 scenes

Sentinel-2 L2A on 116 dates, August 2022 to July 2023

3,050 plots

Detected automatically. The 400 largest were assessed.

Open data

Copernicus imagery via Microsoft Planetary Computer. No credentials, no ground data.

No hyperspectral yet

We first measured what free multispectral data can and cannot do. That result sets the benchmark for the hyperspectral step on the last slide.

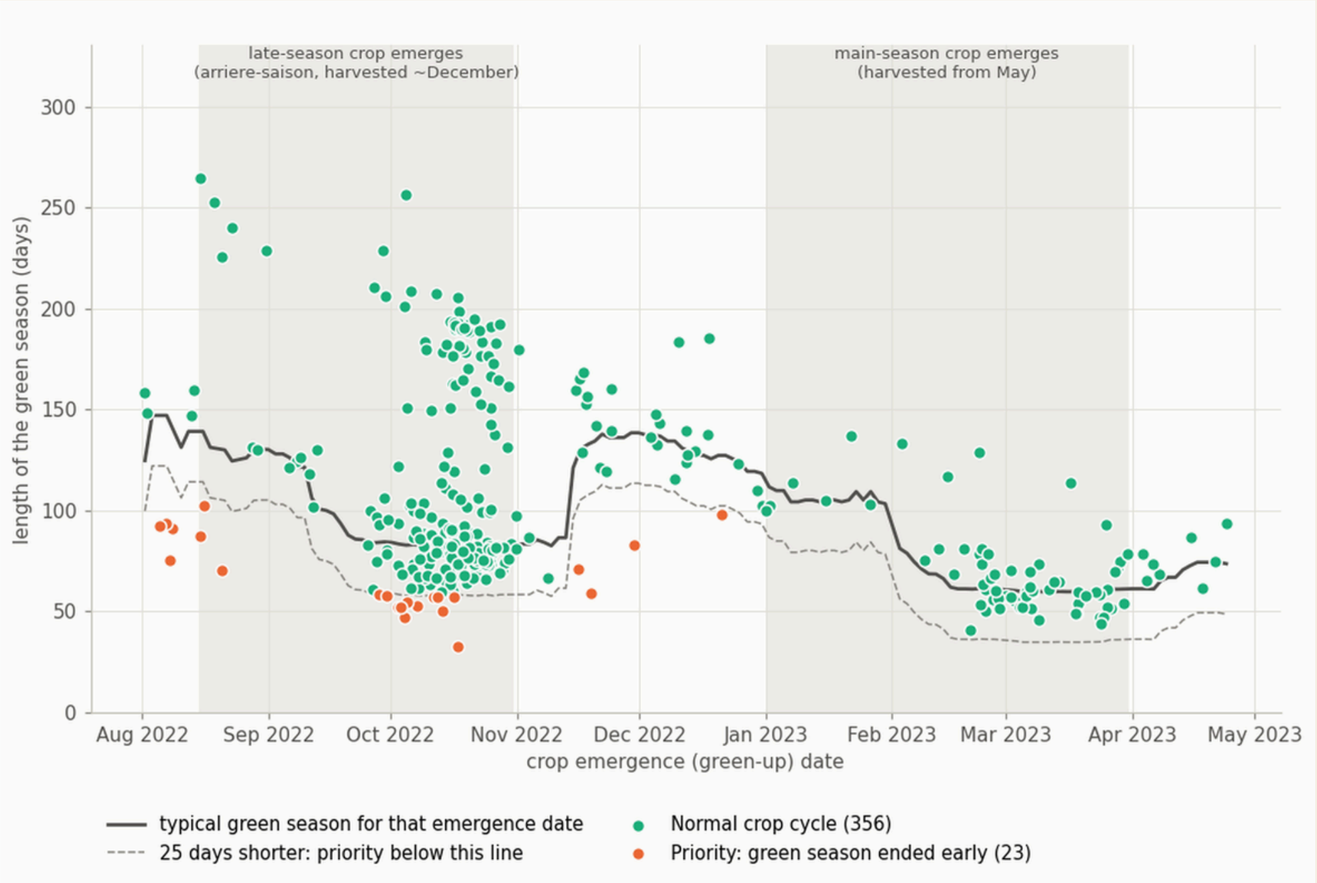
Method

1. Clean. Mask cloud, shadow and cirrus with the SCL band. Compute NDVI (greenness) and NDMI (leaf water).

2. Detect. Pixels that reach NDVI 0.30 once in the year are cultivated. Connected pixels form a plot.

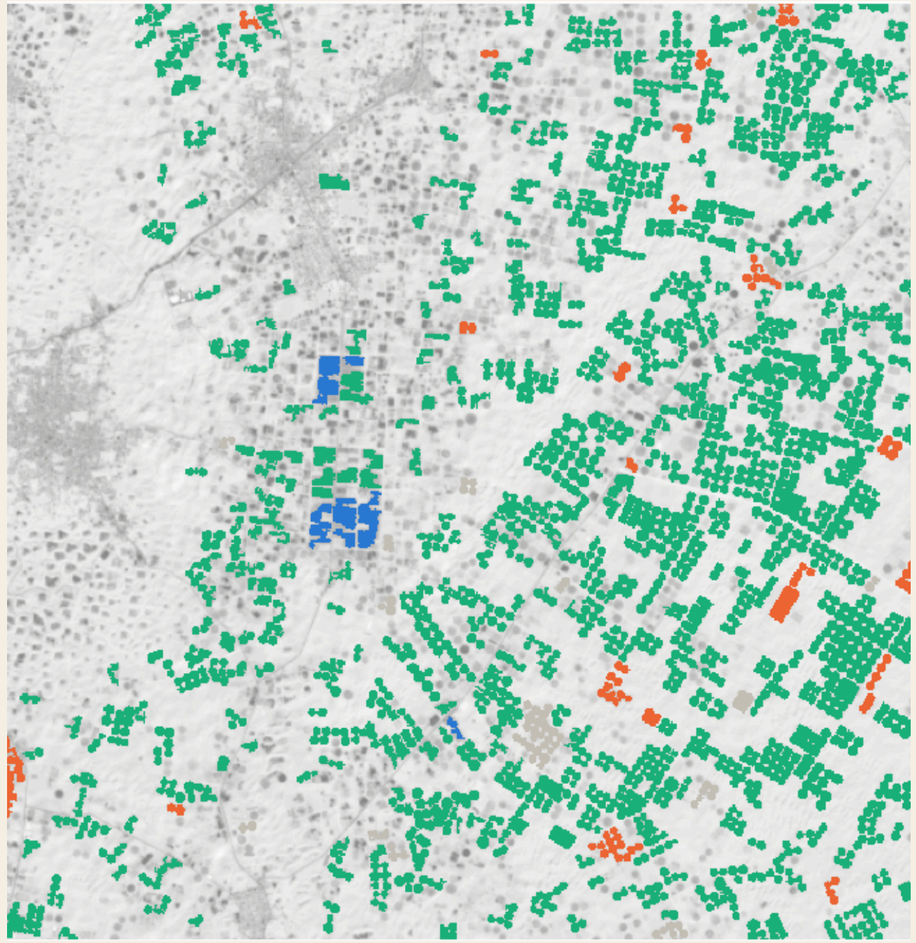
3. Date. Green-up and decline are where a plot crosses 50% of its own seasonal range.

4. Compare. Same green-up window (20 days). Flag a shortfall of 25 days or more.



Each dot is a crop. Orange dots fall below the dashed line and are flagged.

23 plots to inspect first



Orange: priority (23)
Blue: perennial (6)
Green: normal (356)
Grey: not assessable (15)

23 of 379

assessed plots flagged: their green season ended early

58 vs 83.5

days of median green season, flagged plots against normal crops

+38%

cultivated land in the study area, 2020 to 2025 (2,280 to 3,144 ha)

Limitations and validation

An indicator, not proof

No flag has been checked in the field this season. A short season fits water stress, disease or an early harvest.

Tested on a synthetic oasis

44 of 48 planted failures found, none of 208 normal plots flagged. Synthetic data is cleaner than reality, so this is optimistic.

Leaf water adds no stress signal

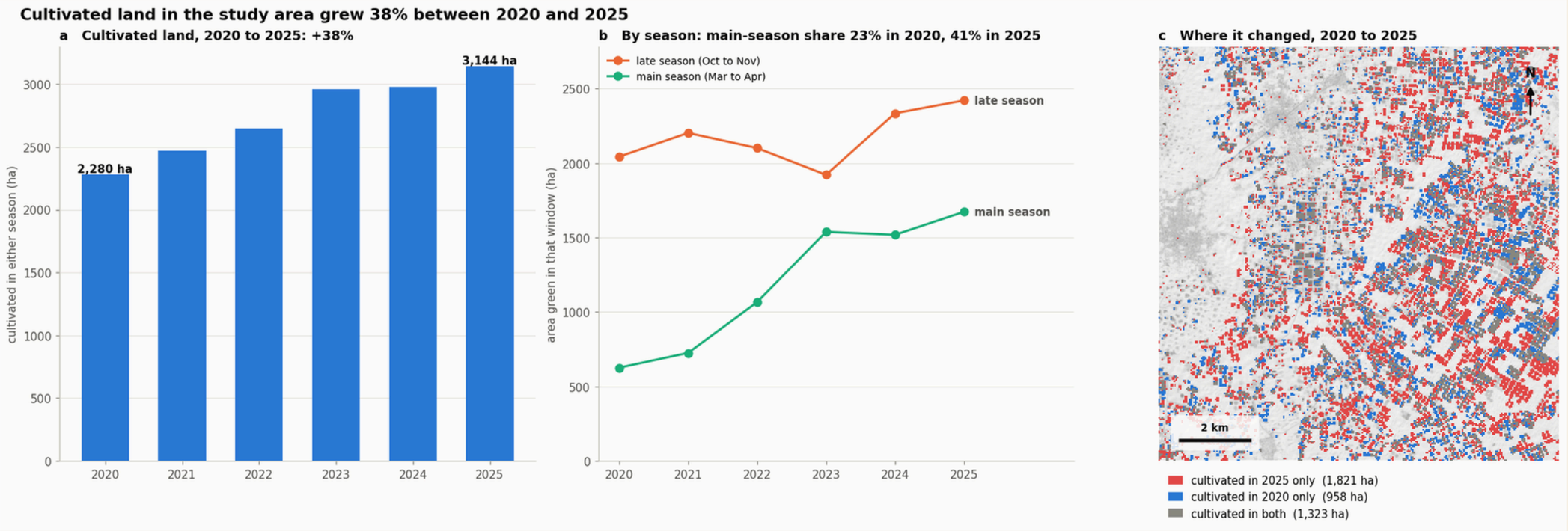
With Sentinel-2 at 20 m, leaf water fell about 3 days before greenness on most crops, about 80%, flagged or not.

Confirmation, not early warning

Replayed with past data only, alarms came about 6 days after the visible greenness decline.

Real validation needs blind field visits to flagged and unflagged plots, or plot-level yield records.

Why it matters: the oasis is changing fast



Cultivated land grew 38% in five years, from 2,280 to 3,144 ha.

The main season's share rose from 23% to 41%. Those crops grow into the hot spring months, when water demand is highest.

The groundwater they draw on has very little modern recharge.

Cultivated land by year, by season, and where it changed between 2020 and 2025.

What we would do next

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1. Validate in the field

Run the pipeline on the 2026 late season. Before the December harvest, visit flagged and unflagged plots without knowing which is which.

2. Test hyperspectral data

Repeat the leaf-water test with narrow bands. We already found an EMIT scene over our area (27 Jan 2023). Next: EnMAP and Arab Satellite 813 (400 to 1700 nm).

3. Refine the method

Split double crops, test the thresholds, use several years as baseline, export plot outlines as GeoJSON for use in the field.

4. Account for water

Combine cultivated area per season with crop water needs to estimate the extra pumping from the move to the main season.

A hyperspectral feature earns its place only if it passes the group test that Sentinel-2 NDMI fails.